



SAINT IGNATIUS' COLLEGE RIVERVIEW

1995 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

MATHEMATICS

3 UNIT (ADDITIONAL) AND 3/4 UNIT (COMMON)

*Time allowed: two hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

Attempt all questions.

There are SEVEN questions. Each question is worth 12 marks.

The mark for each main part of a question is shown in square brackets on the right hand side of the page, eg [2].

All necessary working should be shown. Full marks may not be awarded if work is careless or badly arranged.

Approved calculators may be used. A table of standard integrals is provided.

EACH QUESTION is to be returned in a separate Writing Booklet with the question numbers clearly marked on the cover.

Your examination number must be written on each booklet.

Note: this is Trial Paper only and does not necessarily reflect the content or format of the final Higher School Certificate Examination Paper for this subject.

QUESTION 2. Use a Separate Writing Booklet

- (a) Evaluate $\int_0^2 \sin \theta \cos \theta \, d\theta$ giving your answer correct to 3 significant figures.

[3]

- (b) A particle undergoes simple harmonic motion about the origin O . The displacement, x cm, from O at time t seconds, is given by

$$x = 2 \sin\left(t - \frac{\pi}{3}\right).$$

- (i) Write down the amplitude of the motion.
- (ii) Find the acceleration as a function of time.
- (iii) Express this acceleration as a function of displacement.
- (iv) Find the positive value of t for which the speed is a maximum and determine this speed.

[6]

- (c) The equation $x^3 + 2x - 8 = 0$ has a root close to $x = 1.6$. Use one application of Newton's method to find a better approximation to the root.

[3]

QUESTION 1. Use a Separate Writing Booklet

(a) Solve: $\frac{2}{x-2} < 1$.

[2]

(b) Differentiate with respect to x

(i) $\log \sqrt{2x}$.

(ii) $f(x) = \sec x^3$.

[4]

(c) Evaluate exactly

$$\int_1^{\sqrt{3}} \frac{dt}{\sqrt{4-t^2}}.$$

[3]

(d) (i) Sketch the graph of $y = \cos^{-1} x$.

(ii) Hence or otherwise state the number of solutions to the equation

$$\cos^{-1} x = x.$$

[3]

QUESTION 3: Use a Separate Writing Booklet

(a)

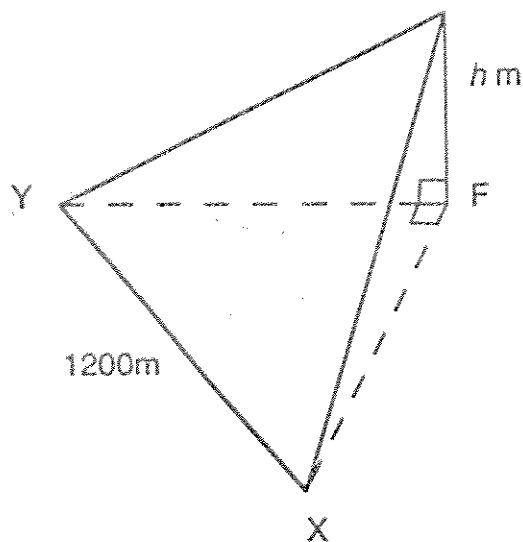


diagram not to scale

Point X is due south and point Y is due west of the foot, F, of a mountain. From X and Y, the angles of elevation of the top of the mountain M are 35° and 43° respectively. If X and Y are 1200 metres apart, show that the height, h metres, of the mountain is given by

$$h = 1200(\tan^2 55^\circ + \tan^2 47^\circ)^{-\frac{1}{2}}$$

Hence evaluate h correct to 2 significant figures.

[4]

- (b) A cube of ice is melting in such a way that it retains the shape of a cube. The surface area of the cube is decreasing at the rate of $10 \text{ cm}^2 / \text{min}$. Find the rate at which the volume is decreasing when the edge is 4 cm.

[4]

- (c) Use the Principle of Mathematical Induction to prove that $13 \times 6^n + 2$ is divisible by 5 for all n , where n is a positive integer.

[4]

QUESTION 4. Use a Separate Writing Booklet

- (a) Evaluate $\int_1^4 t\sqrt{4-t} \, dt$ by using the substitution $t = 4 - u$.

[4]

- (b) Evaluate $\int_0^{\frac{\pi}{4}} \cos x \sin^2 x \, dx$.

[2]

- (c) The acceleration of a particle moving in a straight line is given by

$$\frac{d^2x}{dt^2} = -\frac{72}{x^3}$$

where x metres is the displacement from the origin after t seconds. When $t = 0$ the particle is 9 metres to the right of the origin with a velocity of 4 metres per second.

You may use the result $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$.

- (i) Show that the velocity, v , of the particle, in terms of x , is

$$v = \frac{12}{\sqrt{x}}.$$

- (ii) Find an expression for t in terms of x .

- (iii) How many seconds does it take for the particle to reach a point 35 metres to the right of the origin?

[6]

QUESTION 5. Use a Separate Writing Booklet

- (a) Consider the circle $x^2 + y^2 - 2x - 14y + 25 = 0$.

Show that if the line $y = mx$ intersects the circle in two distinct ~~points~~ ^{points} then

$$(1 + 7m)^2 - 25(1 + m^2) > 0.$$

[3]

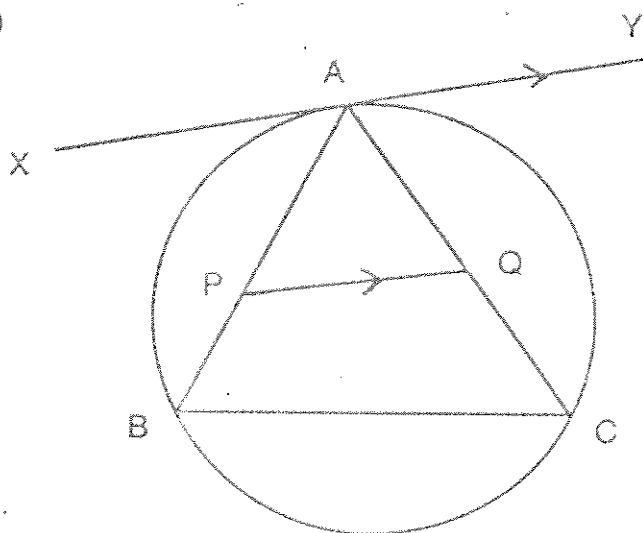
- (b) (i) Show that $x = -4 + Qe^{-3t}$ where Q is a constant, satisfies the equation

$$\frac{dx}{dt} = -3(x + 4).$$

(ii) Describe the rate of change of x with respect to t as t increases without bound.

[3]

(c)



Given $AB = AC$.
The tangent at A is parallel to PQ .

diagram not to scale

Prove

- (i) $AP = AQ$.
- (ii) BC is parallel to the tangent at A .
- (iii) $PQCB$ is a cyclic quadrilateral.

[6]

QUESTION 6. Use a Separate Writing Booklet

(a) (i) Write down the expansion of $\tan(A + B)$.

(ii) Find the value of $\tan\left(\frac{7\pi}{12}\right)$ in simplest surd form.

[4]

(b) Find $\frac{d}{dx} 2\cos^{-1}\left(\frac{x}{4}\right)$.

[2]

(c) During the medieval wars, the enemy wanted to attack a fortress with a 5 metre opening along the front wall. The strategy was to stand at the point P , x metres away from the wall, thus giving an angle of vision α , through which to fire arrows from a cross-bow.

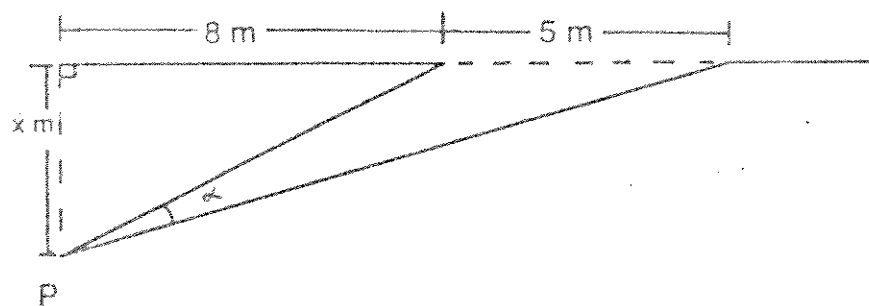


diagram not to scale

(i) Show that the angle of vision α is given by

$$\alpha = \tan^{-1}\left(\frac{13}{x}\right) - \tan^{-1}\left(\frac{8}{x}\right).$$

(ii) Find the value of x in order to give a maximum angle of vision and hence find the maximum angle of vision in radians.

[6]

QUESTION 7.

Use a Separate Writing Booklet

- (a) The roots, α , β and γ of the equation $2x^3 + 9x^2 - 27x - 54 = 0$ are in geometric sequence.

(i) Show that $\beta^2 = \alpha\gamma$.

(ii) Write down the value of $\alpha\beta\gamma$.

(iii) Find α , β and γ .

[5]

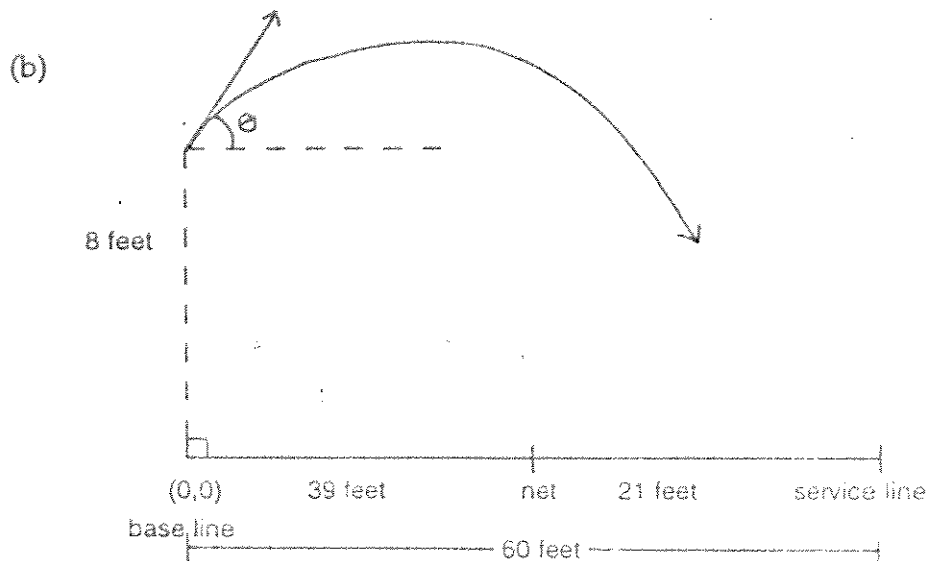


diagram not to scale

In the 1995 Wimbledon Men's Final, "Boom-boom" Becker's serve was measured to have an initial velocity of 184.8 feet/second, or 126 miles/hour. Becker served the ball at the base line from a height of 8 feet at an angle of inclination of θ . In order not to fault, the ball must land within a range of 60 feet.

Taking acceleration due to gravity as 32 feet/second² and the origin as in the diagram,

- (i) derive the equations of motion and show that the position of the ball (x, y) after t seconds is given by

$$x = 184.8t \cos \theta$$

$$y = -16t^2 + 184.8t \sin \theta + 8$$

(ii) Hence show that $y = \frac{-16x^2 \sec^2 \theta}{184 \cdot 8^2} + x \tan \theta + 8$

(iii) Show that if Becker serves the ball horizontally, he will fault.

[7]

END OF PAPER

3 Unit Trial: Riverview: 1995.

1. (a). $\frac{2}{x-2} < 1.$

$$2(x-2) < (x-2)^2.$$

$$x^2 - 4x + 4 - 2x + 4 > 0.$$

$$x^2 - 6x + 8 > 0.$$

$$(x-2)(x-4) > 0$$

$$x > 4 \text{ or } x < 2.$$

(b). (i) $y = \log(2x)^{1/2}$
 $= \frac{1}{2} \log 2x.$

$$y' = \frac{1}{2} \cdot \frac{1}{2x} \cdot 2$$

$$= \frac{1}{2x}.$$

(ii) $f(x) = \sec x^3.$

$$= (\cos x^3)^{-1}$$

$$f'(x) = -1 \cdot (\cos x^3)^{-2} \cdot \sin x^3 \cdot 3x^2$$

$$= -3x^2 \cdot \frac{\sin x^3}{(\cos x^3)^2}$$

$$= -3x^2 \cdot \tan x^3 \cdot \sec x^3.$$

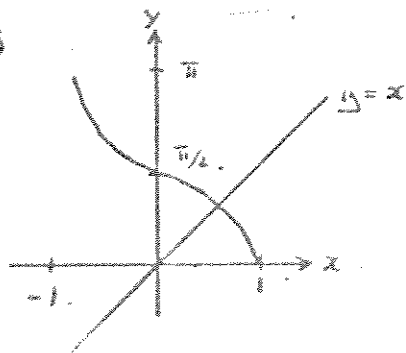
(c) $\int_1^{\sqrt{3}} \frac{dt}{\sqrt{t^2-1}}$

$$= \left[\sin^{-1}\left(\frac{t}{\sqrt{3}}\right) \right]_1^{\sqrt{3}}$$

$$= \sin^{-1}\left(\frac{\sqrt{3}}{\sqrt{3}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{3}}\right).$$

$$= \pi/6.$$

(d) (i)



(ii) one solution.

2. (a). $\frac{1}{2} \int_0^{\pi/2} \sin 2\theta \cdot d\theta.$

$$= \frac{1}{2} \left[-\frac{\cos 2\theta}{2} \right]_0^{\pi/2}$$

$$= -\frac{1}{4} \{ \cos \pi - \cos 0 \}$$

$$= -0.413.$$

(b).

(i) 2

(ii) $\dot{x} = 2 \cos(t - \pi/3)$

$$\ddot{x} = -2 \sin(t - \pi/3)$$

(iii) $\ddot{x} = -x.$

(iv) When $\ddot{x} = 0,$

$$\therefore -2 \sin(t - \pi/3) = 0.$$

$$\sin(t - \pi/3) = 0.$$

$$t - \pi/3 = 0, \pi, \dots$$

$$t = \pi/3, (\pi + \pi/3), \dots$$

$$\dot{x} = 2 \cos(\pi/3 - \pi/3) = 2$$

(c). $x = 1.6 - \frac{f(1.6)}{f'(1.6)}$

$$= 1.6 - \frac{-0.704}{4.68}$$

$$\approx 1.6 + 0.07272$$

$$\approx 1.6727$$

$$[f'(x) = 3x^2 + 2].$$

Q3 (a). $XF = h \tan 55^\circ$

$YF = h \tan 47^\circ$

$XY^2 = YF^2 + XF^2$

$1200^2 = h^2 \tan^2 47^\circ + h^2 \tan^2 55^\circ$

$1200^2 = h^2 (\tan^2 47^\circ + \tan^2 55^\circ)$

$h = \frac{1200}{\sqrt{\tan^2 47^\circ + \tan^2 55^\circ}}$

$= 1200 (\tan^2 47^\circ + \tan^2 55^\circ)^{-1/2}$

$h \approx 670 \text{ m.}$

(b)  $S = 6x^2$
 $\frac{dS}{dx} = -10$

(1) $\frac{dS}{dt} = \frac{dS}{dx} \cdot \frac{dx}{dt}$

$-10 = 12x \cdot \frac{dx}{dt}$

$\frac{dx}{dt} = -\frac{5}{6x}$

(2) $V = x^3$

$\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt}$

$= 3x^2 \cdot \frac{-5}{6x} = -\frac{5x}{2}$

(3) when $x = 4$
 $\frac{dV}{dt} = -\frac{5(4)}{2} = -10$

$\therefore$ Volume decreases at $10 \text{ cm}^3/\text{min}$

(c)(i) When $n=1$

LHS = $13 \times 6 + 2$

$= 80$ which is divisible by 5

$\therefore$ result holds for $n=1$

(2) Assumption: True for $n=k$ $k \in \mathbb{N}$

$\therefore 13 \times 6^k + 2 = 5m$

(3) To prove true for $k+1$

i.e. prove $13 \times 6^{k+1} + 2 = 5p$

LHS = $13 \times 6^{k+1} + 2$

$= 6(13 \times 6^k) + 2$

$= 6(13 \times 6^k) + 12 - 10$

$= 6[13 \times 6^k + 2] - 10$

$= 6[5m] - 10$

$= 5(6m - 2) = \text{RHS.}$

$\therefore$ result is true for $n=k+1$

(4) Since the result holds for an initial value, $n=1$ and if it holds for $n=k$ then it also holds for $n=k+1$,

$\therefore$ result holds for $n=2, 3, 4$ etc for all values of $n \in \mathbb{N}$

$\therefore$ result is true.

Q4. (a).

$\int_3^4 t \sqrt{4-t} dt.$

$= \int_1^0 \frac{v}{(4-v)\sqrt{v}} \cdot \frac{-dv}{v}$

$t = 4-v$
 $\frac{dt}{dv} = -1$
 $dt = -dv$

$= - \int_1^0 4v^{1/2} - v^{3/2} dv$

$= - \left[\frac{8v^{3/2}}{3} - \frac{2}{5} v^{5/2} \right]_1^0$

$= - \left(-\frac{8}{3} + \frac{2}{5} \right)$

$= \frac{34}{15} = 2\frac{4}{15} \checkmark$

(b).

$\int_0^{\pi/4} \cos x (\sin x)^2 dx$

OR
 $u = \sin x$
 $\frac{du}{dx} = \cos x$
when $x=0, u=0$
when $x=\pi/4, u=\frac{1}{\sqrt{2}}$

$= \frac{1}{3} \left[(\sin x)^3 \right]_0^{\pi/4}$

$= \frac{1}{3} \left[\left(\sin \frac{\pi}{4} \right)^3 - (0) \right] = \left[\frac{1}{3} u^3 \right]_0^{\frac{1}{\sqrt{2}}}$

$= \frac{1}{3} \left(\frac{1}{\sqrt{2}} \right)^3 = \frac{\sqrt{2}}{12} \checkmark$

(c).

(i) $-\frac{72}{x^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$\int -72x^{-2} dx = \frac{1}{2} v^2$

$C + 72x^{-1} = \frac{1}{2} v^2$ But $x=9$ when $v=4$

$\therefore C = 0 \checkmark$

$\therefore v^2 = \frac{144}{x}$

$v = \frac{12}{\sqrt{x}} \checkmark$ $v > 0$ only

(ii) $\frac{dx}{dt} = \frac{12}{x^{1/2}}$

$\frac{dt}{dx} = \frac{x^{1/2}}{12}$

$t = \frac{1}{18} x^{3/2} + C \checkmark$

When $t=0, x=9$

$0 = \frac{1}{18} 9^{3/2} + C$

$\therefore t = \frac{1}{18} x^{3/2} - \frac{3}{2} \checkmark$

(iii) $t = \frac{1}{18} 35^{3/2} - \frac{3}{2}$

$\approx 10 \text{ sec.} \checkmark$

Question 5

a) $x^2 + y^2 - 2x - 14y + 25 = 0$

$y = mx$

Solve simultaneously for points of intersection

$$x^2 + m^2 x^2 - 2x - 14mx + 25 = 0$$

$$(1+m^2)x^2 + (-2-14m)x + 25 = 0$$

Two distinct points will occur if there are two solutions to the equation. This occurs when

$$\Delta > 0.$$

$$b^2 - 4ac > 0$$

$$(-2-14m)^2 - 4(1+m^2)25 > 0$$

$$-4(1+7m)^2 - 4 \cdot 25(1+m^2) > 0$$

Divide by -4

$$(1+7m)^2 - 25(1+m^2) > 0.$$

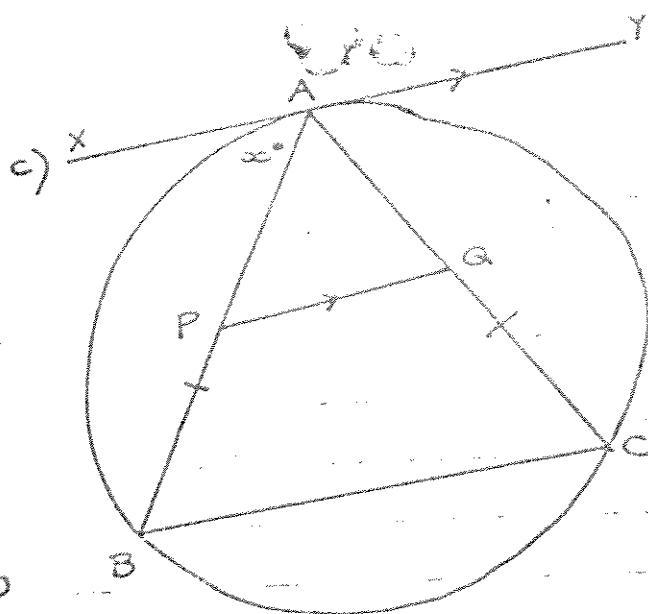
b) i) $x = -4 + Qe^{-3t}$

$$\frac{dx}{dt} = -3Qe^{-3t}$$

$$Qe^{-3t} = x + 4$$

$$\therefore \frac{dx}{dt} = -3(x+4)$$

ii) As $t \rightarrow \infty$ $e^{-3t} \rightarrow 0$



i)

Let $\angle XAB = x^\circ$

$\angle ACB = x^\circ$ (angle in the alternate segment = angle made by tangent and chord)

$\angle ABC = x^\circ$ (base angle of isosceles $\triangle ABC$)

$\angle YAC = x^\circ$ (angle in the alternate segment = angle made by tangent & AC)

$\angle APQ = x$ (alternate angles, equal $XY \parallel PQ$)

$\angle AGP = x$ (alternate angles)

$\therefore AP = AQ$ (sides opposite equal angles are equal)

ii) $\angle XAB = x^\circ = \angle ABC$

(from i)

$\therefore BC$ is parallel to the tangent at A

(alternate angles equal on parallel lines).

(iii) $\angle BPQ = 180^\circ - \infty^\circ$
(angles on straight line AB)

$$\angle BPQ + \angle QCB = 180^\circ - \infty^\circ + \infty^\circ = 180^\circ$$

$\therefore$ PQCB is a cyclic quadrilateral (opposite angles are supplementary)

Question 6

a) i) $\tan(A+B)$
$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

ii) $\tan\left(\frac{7\pi}{12}\right)$

$$= \tan\left(\frac{\pi}{3} + \frac{\pi}{4}\right)$$

$$= \frac{\tan \frac{\pi}{3} + \tan \frac{\pi}{4}}{1 - \tan \frac{\pi}{3} \tan \frac{\pi}{4}}$$

$$= \frac{\sqrt{3} + 1}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}}$$

$$= \frac{4 + 2\sqrt{3}}{-2}$$

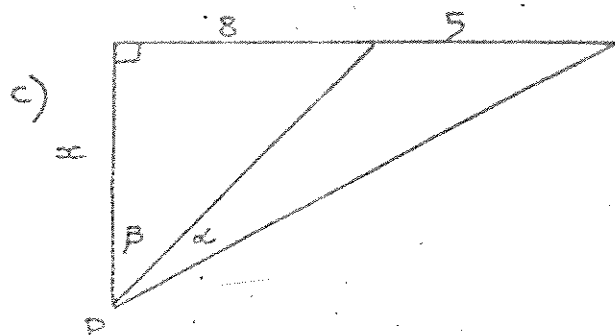
$$= -2 - \sqrt{3}$$

b) $\frac{d}{dx} e^{\cos^{-1}\left(\frac{x}{4}\right)}$

$$= e \times \frac{-1}{\sqrt{1 - \left(\frac{x}{4}\right)^2}} \times \frac{1}{4}$$

$$= -\frac{1}{2} \times \frac{1}{\frac{1}{4}\sqrt{16 - x^2}}$$

$$= \frac{-2}{\sqrt{16 - x^2}}$$



i) $\tan \beta = \frac{8}{x}$

$$\beta = \tan^{-1} \frac{8}{x}$$

$$\tan(\alpha + \beta) = \frac{13}{x}$$

$$\alpha + \beta = \tan^{-1} \frac{13}{x}$$

$$\alpha + \beta - \beta = \tan^{-1} \frac{13}{x} - \tan^{-1} \frac{8}{x}$$

$$\alpha = \tan^{-1} \frac{13}{x} - \tan^{-1} \frac{8}{x}$$

ii) Maximum angle occurs when

$$\frac{d\alpha}{dx} = \frac{1}{1 + \left(\frac{13}{x}\right)^2} x^{-13} x^{-2} - \frac{1}{1 + \left(\frac{8}{x}\right)^2} x^{-8} x^{-2}$$

$$= \frac{\cancel{x^2}}{x^2 + 169} \cancel{x^{-13}} - \frac{\cancel{x^2}}{x^2 + 64} \cancel{x^{-8}}$$

$$0 = \frac{-13}{x^2 + 169} + \frac{8}{x^2 + 64}$$

$$\frac{13}{x^2 + 169} = \frac{8}{x^2 + 64}$$

$$13(x^2 + 64) = 8(x^2 + 169)$$

$$13x^2 + 832 = 8x^2 + 1352$$

$$5x^2 = 520$$

$$x^2 = 104$$

$$x = 2\sqrt{26} \quad (x > 0)$$

Test

x	10	$2\sqrt{26}$	11
$\frac{d\alpha}{dx}$	> 0	0	< 0

$x = 2\sqrt{26}$ gives a maximum value for α .

$$\text{Max. angle} = \tan^{-1} \frac{13}{2\sqrt{26}}$$

$$= \tan^{-1} \frac{8}{2\sqrt{26}}$$

$$\alpha \approx 0.24 \text{ radians.}$$

Question 7

$$a) \quad 2x^3 + 9x^2 - 27x - 54 = 0$$

$$i) \quad \frac{B}{\alpha} = \frac{Y}{B} \quad \text{geometric sequence.}$$

$$B^2 = \alpha Y$$

$$ii) \quad \alpha B Y = -\frac{d}{\alpha} = \frac{54}{2} = 27$$

$$iii) \quad \alpha Y B = 27 \quad B^2 = \alpha Y$$

$$B^3 = 27$$

$$B = 3$$

$$\alpha Y = B^2$$

$$\alpha Y = 9 \quad (Y = \frac{9}{\alpha})$$

$$\alpha + B + Y = -\frac{9}{2}$$

$$\alpha + 3 + \frac{9}{\alpha} = -\frac{9}{2}$$

$$2\alpha^2 + 6\alpha + 18 = -9\alpha$$

$$2\alpha^2 + 15\alpha + 18 = 0$$

$$(2\alpha + 3)(\alpha + 6) = 0$$

$$\alpha = -\frac{3}{2} \text{ or } \alpha = -6$$

$$\text{When } \alpha = -\frac{3}{2}, Y = -6$$

$$\text{When } \alpha = -6, Y = -\frac{3}{2}$$

$$\text{So } \alpha, B, Y \text{ are } -6, 3, -\frac{3}{2} \text{ or } -\frac{3}{2}, 3, -6$$

b) i) Initially

$$\ddot{x}_0 = 0, \quad \ddot{y}_0 = -32 \text{ ft/s}^2$$

$$\dot{x}_0 = 184.8 \cos \theta, \quad \dot{y}_0 = 184.8 \sin \theta$$

$$x_0 = 0, \quad y_0 = 8.$$

$$\ddot{x} = 0$$

$$\dot{x} = C_1$$

when $t=0, C_1 = 184.8 \cos \theta$ when $t=0, \dot{y} = 184.8 \sin \theta$

$$\dot{x} = 184.8 \cos \theta$$

$$\dot{y} = -32t + 184.8 \sin \theta$$

$$x = 184.8t \cos \theta + C_2, \quad y = -16t^2 + 184.8t \sin \theta + C_4$$

when $t=0, x=0$

when $t=0, y=8$

$$\therefore C_2 = 0$$

$$C_4 = 8$$

$$x = 184.8t \cos \theta$$

$$y = -16t^2 + 184.8t \sin \theta + 8$$

ii) Eliminating t .

$$x = 184.8t \cos \theta$$

$$t = \frac{x}{184.8 \cos \theta}$$

$$y = \frac{-16x^2}{(184.8)^2 \cos^2 \theta} + \frac{184.8 \sin \theta \times x}{184.8 \cos \theta} + 8$$

$$y = \frac{-16x^2 \sec^2 \theta}{184.8^2} + x \tan \theta + 8$$

(iii) If ball is served horizontally

then $\theta = 0, \quad y = 0.$

$$0 = \frac{-16x^2}{184.8^2} + 8$$

$$\frac{16x^2}{184.8^2} = 8$$

$$16x^2 = 8 \times 184.8^2$$

$$x = \sqrt{\frac{8 \times 184.8^2}{16}}$$

$$x = 130.7$$

$$x = 131 \text{ feet.}$$

$$> 60 \text{ feet}$$

$\therefore$ Becker will fault.